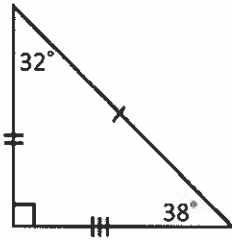


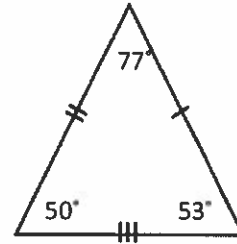
MPM1D – Exam Review – Unit 5 Homework

1. What are the two characteristics that can be used to classify triangles?

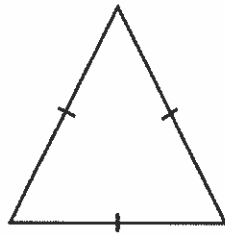
2. Classify the following triangles using the two characteristics from above.



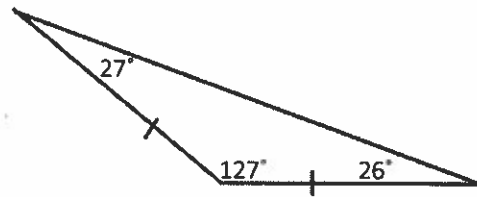
Scalene
Right



Scalene
Acute



Equilateral
Acute



Isosceles
Obtuse

3. Fill in the blanks.

- a. Alternating angles in the Z-pattern have this relationship:

They are equal $\rightarrow$ PLT-Z

- b. The exterior angles in a triangle have a sum of 360° EASTT.

- c. Corresponding angles in the F-pattern have this relationship:

They are equal $\rightarrow$ PLT-F

- d. Angles with a measure between 90° and 180° are called

Obtuse angles.

- e. Complementary angles add up to 90° .

- f. Angles with a measure between 180° and 360° are called

Reflex angles.

- g. Co-interior angles in the C-pattern have this relationship:

They add up to 180° PLT-C

- h. Opposite angles have this relationship:

They are equal OAT

- i. Two or more angles that have a sum of 180° are called

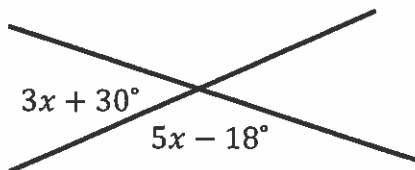
Supplementary angles.

- j. Angles that have a measure between 0° and 90° are called

Acute angles.

4. Solve for x . Show your work and state the theorems that you use to solve. SAT

a.



$$\begin{aligned} 3x + 30 + 5x - 18 &= 180 \\ 8x + 12 &= 180 \\ -12 & \quad -12 \\ \hline 8x &= 168 \\ \frac{8x}{8} &= \frac{168}{8} \\ x &= 21^\circ \end{aligned}$$

b.

$$\begin{aligned} a + 147 &= 180 \\ a &= 33 \end{aligned}$$

SAT

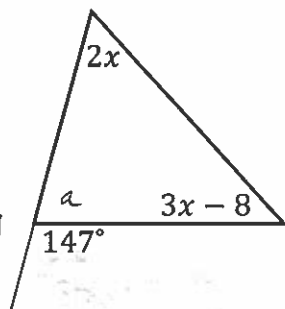
$$a + 2x + 3x - 8 = 180 \quad \text{ASTT}$$

$$33 + 5x - 8 = 180$$

$$5x + 25 = 180$$

$$\frac{5x}{5} = \frac{155}{5}$$

$$x = 31^\circ$$



$$147 = 2x + 3x - 8$$

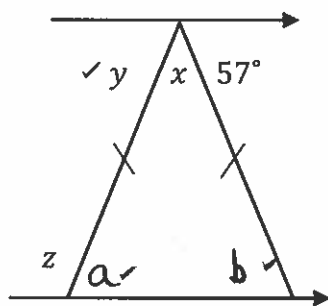
EAT

$$147 = 5x - 8$$

$$\frac{155}{5} = \frac{5x}{5}$$

$$x = 31^\circ$$

5. Determine the measure of each variable below. Show your work.



$$b = 57^\circ \quad \text{PLT-E}$$

$$a = 57^\circ \quad \text{ITT}$$

$$y = 57^\circ \quad \text{PLT-E}$$

$$x + a + b = 180 \quad \text{ASTT}$$

$$x + 57 + 57 = 180$$

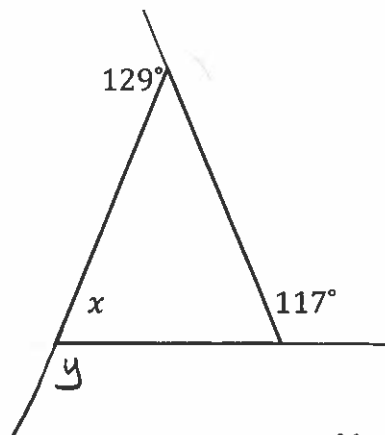
$$x + 114 = 180$$

$$x = 66^\circ$$

$$z + y = 180 \quad \text{PLT-C}$$

$$z + 57 = 180$$

$$z = 123^\circ$$



$$y + 129 + 117 = 360 \quad \text{EASTT}$$

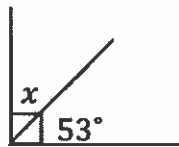
$$y + 246 = 360$$

$$y = 114^\circ$$

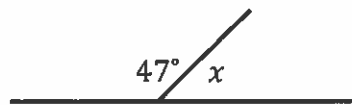
$$x + y = 180 \quad \text{SAT}$$

$$x + 114 = 180$$

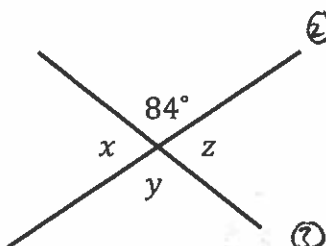
$$x = 66^\circ$$



$$\begin{aligned}
 X + 53 &= 90 && \text{OAT} \\
 -53 &&& -53 \\
 \hline
 X &= 37^\circ
 \end{aligned}$$



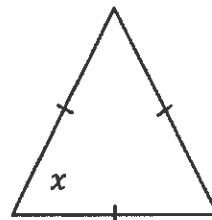
$$\begin{aligned}
 X + 47 &= 180 && \text{SAT} \\
 -47 &&& -47 \\
 \hline
 X &= 133^\circ
 \end{aligned}$$



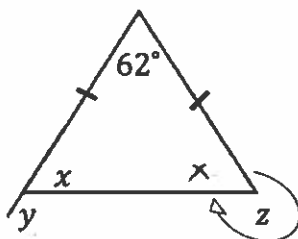
$$\begin{aligned}
 X + 84 &= 180 && \text{OAT} \\
 -84 &&& -84 \\
 \hline
 X &= 96^\circ && \text{SAT}
 \end{aligned}$$

$$\begin{aligned}
 Z &= X && \text{OAT} \\
 Z &= 96^\circ
 \end{aligned}$$

$$\begin{aligned}
 y &= 84^\circ && \text{OAT}
 \end{aligned}$$



$$X = 60^\circ \text{ ETT}$$



$$62 + x + x = 180 \text{ ITT+ASTT}$$

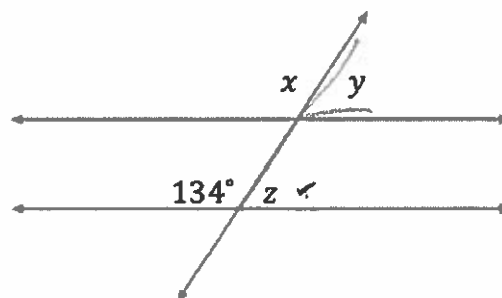
$$\begin{aligned}
 62 + 2x &= 180 \\
 -62 &&& -62 \\
 \hline
 2x &= 118
 \end{aligned}$$

$$\frac{2x}{2} = \frac{118}{2}$$

$$x = 59^\circ$$

$$\begin{aligned}
 x + y &= 180 && \text{SAT} \\
 y &= 180 - 59 \\
 y &= 121^\circ
 \end{aligned}$$

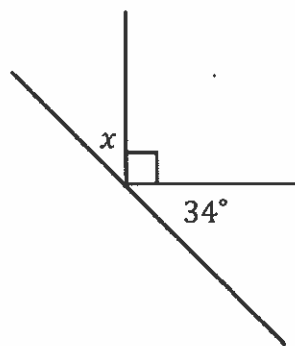
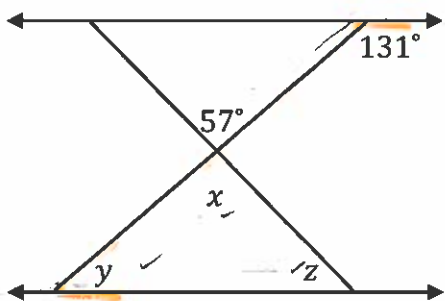
$$\begin{aligned}
 x + z &= 360 \\
 z &= 360 - 59 \\
 z &= 301^\circ \\
 &\text{Reflex angle}
 \end{aligned}$$



$$\begin{aligned}
 z + 134 &= 180 && \text{SAT} \\
 -134 &&& -134 \\
 \hline
 z &= 46^\circ
 \end{aligned}$$

$$\begin{aligned}
 x &= 134^\circ \\
 &\text{PLT-F}
 \end{aligned}$$

$$\begin{aligned}
 y &= z && \text{PLT-F} \\
 y &= 46^\circ
 \end{aligned}$$



$$x = 57^\circ \text{ OAT}$$

$$y + 131 = 180 \text{ PLT-C}$$

$$\begin{matrix} -131 & -131 \\ y & \end{matrix}$$

$$y = 49^\circ$$

$$x + y + z = 180 \text{ ASTT}$$

$$57 + 49 + z = 180$$

$$106 + z = 180$$

$$\begin{matrix} -106 & -106 \\ z & \end{matrix}$$

$$z = 74^\circ$$

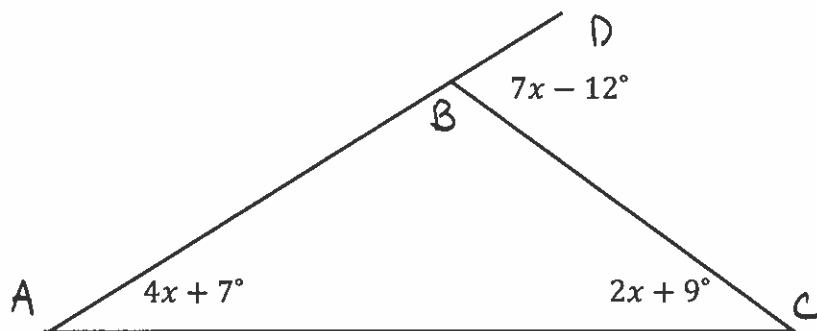
$$x + 90 + 34 = 180 \text{ SAT}$$

$$x + 124 = 180$$

$$\begin{matrix} -124 & -124 \\ x & \end{matrix}$$

$$x = 56^\circ$$

6. Is this angle relationship possible? Provide your reasoning.



$$7x - 12 = 4x + 7 + 2x + 9$$

$$\text{EATT}$$

$$7x - 12 = 6x + 16$$

$$\begin{matrix} -6x & -6x \\ x - 12 & \end{matrix}$$

$$x - 12 = 16$$

$$\begin{matrix} +12 & +12 \\ x & \end{matrix}$$

$$x = 28^\circ$$

$$\begin{aligned} A &= 4x + 7 \\ &= 4(28) + 7 \\ &= 112 + 7 \\ &= 119^\circ \end{aligned}$$

$$\begin{aligned} C &= 2x + 9 \\ &= 2(28) + 9 \\ &= 56 + 9 \\ &= 65^\circ \end{aligned}$$

$$\begin{aligned} B &= 180 - A - C \\ &= 180 - 119 - 65 \\ &= -4^\circ \end{aligned}$$

$\therefore$ This is not possible b/c the triangle cannot have a -ve angle.

Answer Key

1a) Side Lengths and Angles

2a) Scalene, Right 2b) Scalene, Acute 2c) Equilateral, Acute 2d) Isosceles, Obtuse

3a) They are equal 3b) 360° 3c) They are equal 3d) obtuse

3e) 90° 3f) reflex 3g) They add up to 180°

3h) They are equal 3i) supplementary 3j) *Awk .*

4a) $x = 21^\circ$ SAT 4b) $x = 31^\circ$ EATT or SAT + ASTT

5a) $x = 66^\circ, y = 57^\circ, z = 123^\circ$ PLT-2, ITT, PLT-4, ASTT 5b) $x = 66^\circ$ EASTT + SAT

5e) $x = 37^\circ$ CAT 5f) $x = 133^\circ$ SAT

5g) $x = 96^\circ, y = 84^\circ, z = 96^\circ$ SAT + OAT 5h) $x = 60^\circ$ ETT

5i) $x = 59^\circ, y = 121^\circ, z = 301^\circ$ ITT, SAT, ASTT 5j) $x = 134^\circ, y = 46^\circ, z = 46^\circ$ SAT, PLT-P

5k) $x = 57^\circ, y = 49^\circ, z = 74^\circ$ OAT, PLT-C, ASTT 5l) $x = 56^\circ$ SAT

6) $x = 28^\circ$ EATT No it is not possible, the sum of the angles A, B, and C is greater than 180°